



Balancing the workload between digital and human actors in Knowledge Work

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Vincent Antonio Balau

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Abstract

We have heard hundreds of times that Artificial Intelligence (AI) will be the next big revolution. Economic reports from McKinsey and Boston Consulting show that companies are struggling with AI integration. How can companies and organizations in the knowledge work sector reliably integrate AI tools in a way that brings value? This thesis analyzes nine core papers with the aim of identifying decision factors for delegating tasks to AI agents. In other words, precisely those factors that should be considered when deciding, within a work process, which tasks AI should take over and which tasks require human control. The study identifies the following four decision factors: the inherent structure of a task, the nuances and context of a task, the potential impacts of a task, and the additional overhead for the management and coordination of human-machine work processes. In addition, five general design guidelines were developed: defining process intention, selecting the degree of automation, calibrating explainability to risk, managing trust between agents, and redesigning human roles. These design guidelines are primarily intended to serve as frameworks for the development and refinement of work processes. Overall, the results serve as conceptual pillars for the integration of AI tools into corporate and organizational processes.

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Introduction

The promise of Artificial Intelligence (AI) as a driver of business value is hard to ignore. Consulting reports from Boston Consulting Group (BCG) [1] and McKinsey [23] describe AI as one of the most impactful technologies of our time. Yet the same reports reveal that most companies struggle to generate measurable value from their AI investments. BCG's 2025 report found that 60% of companies reported minimal gains despite substantial AI investment, while McKinsey's survey of nearly 2,000 executives reports a similar pattern, with only around 6% of organizations capturing meaningful enterprise value from AI, even though 88% claim to use it regularly. The majority of organizations invest in AI without a clear understanding of where it creates value. These reports attribute the problem primarily to strategic and organizational failures, blaming the lack of C-level leadership in technological adoption. While these observations are important, they operate on the level of organizational strategy. This thesis focuses on a more specific problem. Before an organization can lead AI adoption effectively, it needs to know which tasks are suitable for AI in the first place. Many organizations remain uncertain about what to hand over to AI, which processes are suitable for automation, and where the boundaries of responsible AI integration lie. This leads to the central question: "Which tasks should be handed to AI, and which should remain under human control?" To address this question, this thesis uses a structured literature review (SLR), applying the PICOC framework to define the search scope [17]. It synthesizes current research to provide an overview of the engineering guidance available for redesigning business processes with AI in knowledge work. This leads to two guiding research questions:

- RQ1:** What principles and decision factors does current research identify for allocating tasks between AI and human actors in knowledge business processes?
- RQ2:** How can the found principles be applied or extended to help (re)design work processes for responsible and reliable AI integration?

The central findings are organized around four key decision factors that can be used to assess whether and how a task can be allocated to AI. The first is the *structural logic of a task* and represents the degree to which its internal logic can be algorithmically formalized. The second is the *interpretive and contextual nuance* and describes to what extent a task depends on forms of understanding that are difficult to derive from data alone, such as social dynamics, emotions, and ethical judgments. The third is *risk and accountability* and analyzes the potential consequences of an incorrect AI decision for the stakeholders involved. The fourth is called *multi-agent collaboration dynamics* and assesses the overhead introduced when human agents and AI agents must work together on a shared task. From these decision factors, a set of design guidelines is derived that addresses the problems of a task analyzed through the four decision factors. These guidelines cover the suggested degree of automation, the required level of explainability and fairness monitoring, the architecture of human control, and the management of trust between human and AI agents.

This thesis involves a structured literature review conducted in the context of a bachelor's thesis. The thesis synthesizes existing research rather than gathering fresh empirical data to validate its findings. The literature review is grounded in a curated set of nine core papers, selected via a structured process. The findings present a focused synthesis rather than an exhaustive survey of the field.

The thesis is structured as follows. Chapter 2 introduces the theoretical foundations and reviews related work, embedding the research in the context of business process management and knowledge work. Chapter 3 describes the structured literature review process, including the PICOC framework, the search strategy, and the selection criteria applied. Chapter 4 presents the core analysis, identifying the four key decision factors and deriving the design guidelines for responsible AI integration. Chapter 5 critically reflects on the findings, discusses limitations of the current literature, and addresses open conceptual questions. Chapter 6 summarizes the answers to the two research questions and outlines directions of future research.

Related Work and Theoretical Foundations

2.1 Business Process Management and Process Redesign

The question of which tasks to allocate to AI requires a structured way of describing work. Business Process Management (BPM) provides both vocabulary and the analytical framework for this purpose. BPM is the research field where identifying, modeling, improving, analyzing, designing, and monitoring business processes are the main focus of research. A business process is defined as “[...] a collection of interrelated events, activities, and decision points that involve a number of actors and objects, which collectively lead to an outcome that is of value to at least one customer.” [12]. Dumas et al. [12] define the term “task” as a fine-grained singular unit of work that is processed by a single process participant. The term “activity,” in contrast, refers to a more coarse-grained idea of work units. This vocabulary provides the unit of analysis for this thesis. However, this definition assumes that a task is executed by a single participant. The frameworks reviewed for this thesis frequently propose modes of collaboration in which human and AI agents jointly contribute to the same task. A common example is when AI makes a recommendation and a human validates or overrides it. Section 2.3 examines the different granularity levels found in literature and defines the unit of analysis used throughout this thesis. Within the BPM life cycle introduced by Dumas et al. [12], the problem of task allocation belongs to the process redesign phase. The main objective is to identify changes that would help to address the issues identified in the process discovery and analysis phase to improve the process and meet performance objectives.

A foundational concept in several of the reviewed sources is Task-Technology Fit (TTF), introduced by Goodhue and Thompson [16], which holds that technology produces performance gains only when capabilities align with the demands of the task it supports. Engel et al. [13] build on this foundation by extending the unit of analysis from the

individual task to the broader process context, arguing that fit must be evaluated not only on the task level but also in relation to organizational and social requirements of the process. This extension is directly relevant for the allocation framework developed in Chapter 4, where task characteristics are always assessed in combination with contextual and risk factors, rather than in isolation.

2.2 Knowledge Work and Cognitive Automation

A further restriction on the scope of this thesis concerns the type of work under consideration. The focus is specifically on knowledge work, defined as work that is intellectual, creative, non-routine, and involves the application of knowledge [9]. Knowledge workers typically operate in fields like medicine, law, consulting, and education, where expertise is acquired through training and experience is central to the work [9].

Previous waves of automation, like workflow management (WfM) and robotic process automation (RPA), targeted rule-based, routine tasks with few exceptions and little to no cognitive reasoning [13]. Both approaches operate on deterministic logic and well-defined rules. With the emergence of AI systems capable of cognitive functions such as natural language understanding and pattern recognition, automation now extends to tasks that do not require full formalization to be processed. Engel et al. [13] describe this shift as cognitive automation, which moves beyond rule-based processing toward probabilistic outcomes based on machine learning.

A central concept in this context is tacit knowledge, a term attributed to Polanyi [21], who used it to describe all the knowledge that is difficult to articulate. Knowledge workers rely on tacit knowledge due to the vast amounts of unstructured data they deal with from various sources [30]. The challenge of tacit knowledge is not only that it is hard to transfer to AI systems, but it is sometimes impossible to codify at all. Autor [2] applied this insight directly to automation, arguing that tasks requiring adaptability, contextual judgment, and common sense resist machine substitution because their underlying knowledge cannot be formalized. This constraint has direct implications for the allocation of tasks between humans and AI, which is explored in Chapter 4.

The task allocation problem for knowledge work becomes a question of whether a task can be represented in a form an AI system can process reliably and to what extent non-technical factors like social dynamics and contextual judgment remain beyond the system's reach. Before this allocation logic can be applied, the unit of analysis must be clearly defined.

2.3 Defining the Unit of Analysis

Section 2.1 defined a task in the BPM sense as a fine-grained unit of work processed by a single participant. The reviewed literature views this concept at different granularity levels. This section surveys the different perspectives because the level of granularity at which the task is defined directly affects the allocation decision.

The most granular approach decomposes a task into cognitive sub-processes. Rather than treating tasks as uniform blocks of work, this view aligns specific AI competencies with discrete forms of human cognition. In the domain of predictive maintenance (PdM), Stohr and O'Rourke [24] decompose a task into sets of cognitive functions that represent the mental capabilities of a human brain, including all life functions and mental processes. These may encompass fundamental subconscious human behaviors, such as memory and perception, as well as conscious, more intricate functions, including recognition and problem-solving. They introduced the key distinction between core cognitive functions (CCF), in which AI systems inherently outperform humans due to the mathematical and data-intensive nature of these functions. Symbiotic cognitive functions (SCF), on the other hand, which involve ambiguity and subjective preference, require collaboration between human and AI agents, rather than substitution.

In a similar approach, Xiang and Yu [29] decompose a task into mental sub-steps. Tasks are decomposed into four sequential layers: an intention layer where a goal is defined, a reasoning layer where the data is analyzed, a decision layer where conclusions are reached, and an action layer where outcomes are executed. While this aligns with the focus on mental processing proposed by Stohr and O'Rourke [24], Xiang and Yu [29] describe how agency shifts between agents at each layer. By isolating these functions, the authors provide a framework where AI is not seen as replacing a whole task but as substituting specific mental capabilities and augmenting others. In addition, Schroder et al. [22] use the model of cognitive functions for their approach, suggesting that this granular analysis of tasks helps decide which aspects of a task can be automated.

A complementary approach is offered by Vimalkumar et al. [26], who classify tasks in a hierarchy, ranging from simple and problem tasks, which have determined outcomes and clear path solutions, to decision, judgment, and fuzzy tasks, which involve multiple competing outcomes, probabilistic interdependencies, and multi-dimensional models. This taxonomy provides a practical basis for assessing the degree to which a task can be algorithmically formalized and thus determining the appropriate mode of automation.

At a more formal level, Dubois and Le Ny [11] model a task as a binary decision problem, where each unit of work requires a choice between a null hypothesis and a signal event, assigned either to a human or AI agent depending on the current trust and workload state of the human operator. Each new task is independent of previous ones and requires a decision to be made by either human workers or AI agents. It must be stated that these assumptions about the task do not fit the usual environment of knowledge work. While the model by Dubois and Le Ny [11] excels in high-frequency, repetitive classification environments, knowledge work is typically characterized by

iterative processes, ambiguity, and synthesis of unstructured information. Nevertheless, an important concept is introduced: the idea that the optimal task assignment can dynamically change based on the current workload and trust state of the human agent.

Zhang et al. [31] take a coarser view, treating tasks as discrete units of work, derived from a larger project, each with specified resource requirements. This implies that tasks are well-defined, with quantifiable needs for execution.

Bidar et al. [4] define a task as a set of activities. Similar to Stohr and O'Rourke [24], even those activities can be broken down further into mindless routine components and those requiring mindful judgment. This distinction suggests substituting the mindless routine components, while judgment-dependent components require either human involvement or carefully designed augmentation architecture. A shift toward more coarse-grained scopes is evident in the work of Engel et al. [13], who identify automation potential on the use case level, which encompasses business processes rather than individual cognitive functions or entire jobs. It has the purpose of focusing on business value creation and linking it to organizational business requirements and context.

The reviewed perspectives range from sub-task-level cognitive functions (Stohr and O'Rourke [24]) to entire use cases (Engel et al. [13]). This range reflects a central tension in the literature, where finer granularity enables more precise allocation decisions but risks losing sight of the broader context in which the task is embedded. The goal of the thesis is to assess the allocation of individual tasks rather than whole processes. Only when work is available at this granular level can it be assessed whether it can be reliably processed by an AI system. An atomic, well-defined task can be immediately assessed for human or AI assignment. A coarser task that involves multiple sub-tasks cannot be allocated as a whole. It must first be decomposed before any meaningful allocation decision becomes possible. However, a fine-grained view alone is not sufficient. The broader perspective on a task is equally important, as certain factors only become visible at a higher level of analysis. The consequences of a wrong decision, the impact on stakeholders, or the ethical implications of an automated outcome cannot be assessed by looking at an isolated sub-task. They require an understanding of the larger process in which the task is embedded. Therefore, effective task allocation requires both perspectives. The fine-grained view enables precise allocation of individual sub-tasks, while the broad view ensures that critical factors such as risk and accountability are not overlooked. These two perspectives are applied throughout the analysis in Chapter 4. This is consistent with Engel et al. [13], who similarly argue that automation potential must be evaluated in relation to the organizational context of the process.

For the purpose of this thesis, a task is therefore understood as the smallest meaningful unit of work within a process, while acknowledging that its execution may involve the coordinated contribution of more than one agent.

Methodology

3.1 Research Design

This thesis performs a structured literature review process to synthesize existing research to provide a focused overview of the current guidance for constructing AI systems and task allocation between human and AI actors. This includes the definition of the search space via the PICOC framework [20], the documentation of the multi-stage search strategy (including semantic discovery and database queries), and the application of explicit inclusion and exclusion criteria for the selection of core research papers. The goal is a qualitative synthesis where recent literature is evaluated and core findings are presented.

3.2 Definition of the Search Space

To define the scope of the systematic literature review and derive relevant search terms and robust research questions, the PICOC framework (Population, Intervention, Comparison, Outcome, and Context) was applied. Originally rooted in social sciences [20], this framework was specifically adapted for the field of Software Engineering by Kitchenham and Charters [17] to ensure a structured and rigorous approach to evidence-based research.

The five dimensions of the framework are defined as follows. Population identifies the target subjects or core problem area. Intervention specifies the core approach or methodology under investigation. Comparison describes the alternatives against which the intervention is assessed. Outcome determines the specific results or types of evidence the study aims to measure. Context defines the setting or organizational circumstances of the research. The application of this framework to both research questions is shown in table 3.1 and served as the basis for deriving the search strings for database querying.

Table 3.1: Application of the PICOC framework to the research questions.

Components	RQ1: Identification of key principles	RQ2: Application and Design
Population (P)	Human workers and AI agents are involved in knowledge-intensive business processes.	Identical to RQ1.
Intervention (I)	Approaches and frameworks for allocating or balancing tasks and decision-making responsibilities between human and AI actors.	Adapting identified principles into actionable guidelines or frameworks for responsible AI-driven process (re)design.
Comparison (C)	Purely human task execution, fully automated (AI-agents only) task execution, and hybrid human-AI collaboration settings.	Identical to RQ1
Outcome (O)	Identified principles, decision factors, frameworks, or best practices guiding task allocation between human and digital actors.	Frameworks or design recommendations that enable reliable AI integration into organizational processes.
Context (C)	Knowledge-intensive organizational and business processes, where human expertise, experience, or contextual judgment are critical to achieving process outcomes.	Identical to RQ1.

3.3 The Literature Search Process

The search process followed three stages: expert-guided seed paper identification, AI-supported semantic discovery, and systematic database searches. The first stage followed a “seed paper” strategy, initiated through expert consultation with the thesis supervisor. This stage led to the identification of Bidar et al. [4] as a foundational work, which provided an initial framework for approaching the research questions from a risk-informed perspective, addressing both social risk and technical task complexity.

While Bidar et al. [4] provided a solid foundation for risk-informed design, a single source was insufficient to capture the broad research landscape required for this study. To broaden the conceptual scope beyond the initial seed paper and mitigate the risk of citation bias, the “Find Topics” feature of the AI-supported research platform SciSpace was utilized. This tool facilitates semantic discovery by processing natural language queries to generate four primary outputs: (1) comprehensive definitional overviews, (2) semantically linked academic concepts, (3) curated literature lists, and (4) cross-disciplinary application contexts. The primary query used was, “What are the key decision factors that influence the allocation of tasks between AI and human actors in knowledge-intensive business processes?” Results were documented in a research notebook. As the

internal search logic of SciSpace is not fully transparent, this stage should be understood as a complementary discovery step rather than a reproducible systematic search. It was included as a deliberate methodological choice to extend the search space beyond the limitations of keyword-based database queries. This process identified recurring themes in literature, including task complexity and its relation to automation potential, the role of contextual and tacit knowledge as a barrier for automation, and risk-informed approaches to task allocation. These themes informed the search strings used in the subsequent database searches and provided an initial orientation for the analysis in Chapter 4.

The third stage consisted of systematic searches across three academic databases: IEEE Xplore, ACM Digital Library, and AIS Electronic Library. The search strings were derived from the PICOC dimensions and iteratively refined to balance precision and recall. All searches were restricted to recent publications, with IEEE covering 2020 to 2026 and ACM and AIS covering 2019 to 2026. All database searches were conducted on January 9, 2026. Results were filtered to include only journal articles and conference papers.

IEEE Xplore

```
((("Document Title":"Task Allocation in Human Machine" OR "Document Title":"Task-
Allocation" OR "Document Title":"Process Automation" OR "Document Title":"Workflow
Automation" OR "Document Title":"Human-machine Hybrid Augmented Intelligence")
AND ("Abstract":"AI" OR "Abstract":"artificial intelligence" OR "Abstract":"agents"
OR "Abstract":"Human Machine") AND ("Abstract":"knowledge-intensive" OR "Ab-
stract":"business process" OR "Abstract":"workload" OR "Abstract":"task allocation")
NOT "Document Title":"robotic" NOT "Document Title":"robot" NOT "Document Ti-
tle":"robots"))
```

ACM Digital Library

```
[[Abstract: "design guidelines"] OR [Abstract: "design principles"] OR [[Abstract: "frame-
work"] AND [Abstract: "conceptual model"]]] AND [[Abstract: "artificial intelligence"]
OR [Abstract: "ai"] OR [Abstract: "generative ai"] OR [Abstract: "llm"]] AND [[Ab-
stract: "business process"] OR [Abstract: "knowledge work"] OR [Abstract: "workflow
management"]] AND NOT [Abstract: "robot"] AND NOT [Abstract: "uav"] AND NOT
[Abstract: "sensor"] AND NOT [Abstract: "hardware"] AND NOT [Abstract: "edge
computing"] AND NOT [Abstract: "battery"] AND [E-Publication Date: (01/01/2019
TO 02/28/2026)]
```

AIS Electronic Library

```
("Human-AI" OR "Hybrid Intelligence") AND "Task Allocation" AND "Business Process"
```

The IEEE query targeted document titles containing terms related to task allocation, process automation, and human-machine collaboration, combined with abstract-level

filters for AI and business process contexts. Publications related to physical robotics were excluded. The search was restricted to publications from 2020 to 2026 and returned 97 results. The ACM query focused on abstracts containing terms related to design guidelines, frameworks, and conceptual models in the context of AI and business processes, with exclusion for hardware-related domains such as sensors, UAVs, and edge computing. The search was restricted to publications from 2019 to 2026 and returned 10 results. The AIS Electronic Library query combined terms for human-AI collaboration, hybrid intelligence, task allocation, and business processes. The search was restricted to publications from 2019 to 2026 and returned 8 results. The combined database searches resulted in 115 results across the three databases.

3.4 Selection Criteria

The following inclusion and exclusion criteria described in table 3.2 were applied to filter the search results.

Table 3.2: Inclusion and Exclusion Criteria

Criteria	Description	Selection
Period	Relevant time period of published articles	Include: 2019–2026 Exclude: prior to 2019
Language	Language of the publication	Exclude: non-English articles
Publication Type	Type of publication	Include: peer-reviewed journal articles or conference papers
Accessibility	Availability in university databases	Exclude: not accessible
Relevance to RQs	Articles excluded if not relevant to a research question	Exclude: not relevant to at least 1 RQ

3.5 Selection Process

The selection criteria were applied to the 115 results from the database searches. Papers were screened based on the title and abstract to assess their relevance to at least one of the two research questions. Papers clearly outside the scope, such as those focused on physical robotics, sensor networks, or purely technical benchmarks without an allocation or design perspective, were excluded.

From the 115 initial results, 14 papers were selected for full-text review: 4 from IEEE, 3 from ACM, and 7 from AIS Electronic Library. After full-text reading, 6 papers were retained. The remaining 8 were excluded because they did not directly address the task allocation problem or did not provide frameworks applicable to knowledge business processes. For example, Benzin and Rinderle-Ma [3] focused on event prediction methods rather than allocation decisions, Wallace et al. [27] addressed RPA implementation

without a task allocation perspective, and Fahse and Schmitt [14] concentrated on sales forecasting without generalizable design principles.

The six papers retained from the database searches were supplemented by three additional papers identified through other channels. Bidar et al. [4] was identified as a seed paper through expert consultation with the thesis supervisor, as described in Section 3.3. Engel et al. [13] and Vimalkumar et al. [26] were identified during the semantic discovery phase using SciSpace. These three papers were included because they addressed core aspects of the research questions, specifically cognitive automation assessment, task complexity and algorithmic fairness, and human-AI symbiosis in process automation that were not sufficiently covered by the papers found through the database searches alone. The final corpus consists of nine core papers listed in table 3.3.

Table 3.3: Core papers selected for analysis.

Author	Year	Primary Focus
Bidar et al. [4]	2023	Risk-centered task allocation in the public sector
Dubois and Le Ny [11]	2020	Adaptive allocation based on trust and workload
Engel et al. [13]	2024	Assessing use case amenability to cognitive automation
Schroder et al. [22]	2022	Human-AI coordination in digital service companies
Stohr and O’Rourke [24]	2021	Cognitive functions lens for predictive maintenance
Vimalkumar et al. [26]	2021	Task complexity, opacity, and algorithmic fairness
Xiang and Yu [29]	2023	Human-machine hybrid augmented intelligence
Yun et al. [30]	2025	Generative AI in knowledge work and data navigation
Zhang et al. [31]	2025	Role-based task allocation in multi-agent environments

3.6 Snowballing Method

To complement the database searches and capture relevant foundational work not indexed under the search terms used, backward snowballing was applied. The reference lists of the selected papers were reviewed for additional relevant sources. This process was selective rather than exhaustive. When a reviewed paper built on a specific theoretical concept, the original source was traced and assessed for inclusion. For example, the reference to Task Technology Fit by Goodhue and Thompson [16] was identified through Engel et al. [13], Polanyi [21] was traced through Autor [2], and the task complexity typology by Campbell [7] was identified through Vimalkumar et al. [26]. No systematic forward snowballing was conducted, as the scope of a bachelor thesis limited the feasible search effort. The final corpus of nine core papers, supplemented by foundational references identified through backward snowballing, forms the basis of the analysis in Chapter 4.

3.7 Synthesis and Curation Method

The selected papers were analyzed using a synthesis matrix. Each paper was mapped against six categories derived from the research questions: task and context definition, task-related factors, human-related factors, AI and system-related factors, allocation logic, and design principles. The first category captured how each paper defined its unit of analysis and the domain in which it was applied. The next three categories cover the inherent properties of the task, the constraints and capabilities of the human agent, and the characteristics of the AI system. The allocation logic category documented how each paper translated its factors into assignment decisions. The design principles category captured any actionable recommendations for system or process design.

This matrix served two purposes. First, it enabled a structured comparison across papers, making it possible to identify where findings converged and where they diverged. Second, it provided the organizational basis for the four key decision factors presented in Chapter 4. A recurring observation during the synthesis was that the three factor classes (task, human, AI) did not function in isolation. Changes in one factor often triggered shifts in the requirements of another. This interdependency led to the decision to organize the analysis in Chapter 4 around four cross-cutting decision factors rather than around the three separate factor classes. The complete synthesis matrix is provided in Appendix A in the table A.1.

Analysis of Task Allocation and System Design

The findings from the literature review are synthesized in this chapter, with the purpose of addressing the core research questions. In the first half of this chapter, key decision factors influencing task allocation between human and digital actors are identified and categorized, directly addressing research question RQ1. The identified factors then lead to different allocation logic approaches. Building on this foundation, the second half of this chapter (Section 4.6) outlines the design guidelines for responsible and reliable AI integration into work processes, addressing research question RQ2.

4.1 Key Decision Factors for Task Allocation

From the synthesis of the reviewed literature, four key decision factors have emerged that collectively inform the allocation decision. Each decision factor represents a distinct analytical perspective on a task, and each brings its own set of factors and considerations. The first decision factor examines the structural logic of a task, including its complexity and cognitive requirements. The second shifts the focus to interpretive and contextual nuance, assessing the degree to which a task depends on tacit knowledge and inherently human capabilities such as moral judgment. The third decision factor addresses risk and accountability, evaluating the consequences of failure and the need for human oversight. The fourth considers the dynamics of multi-agent collaboration, examining the frictions that arise when human and AI agents work together on a shared task. The following sections analyze each of these decision factors in detail.

4.2 Structural Logic of a Task

The first decision factor for task allocation in knowledge work is the inherent structural logic of a task. This decision factor evaluates the extent to which a task's internal logic can be mathematically or algorithmically formalized. The general idea is to assess whether the structure of a given task allows for reliable automation.

4.2.1 Task Complexity

The theoretical framework introduced by Vimalkumar et al. [26] assesses the feasibility of automating a task based on its complexity while also considering the opacity of algorithms and resulting implications for algorithmic fairness. The task complexity is classified into five categories influenced by the theories of Campbell [7] and Wood [28]. These five types are (with increasing complexity):

1. **Simple Task:** clearly defined path to a single desired outcome (e.g., interest calculation)
2. **Problem Task:** multiple paths that can lead to a single desired outcome (e.g., route optimization)
3. **Decision Task:** multiple paths that can lead to multiple possible outcomes (e.g., recommendation systems)
4. **Judgment Task:** process multiple pieces of information from multiple sources to predict future outcomes (e.g., credit scoring)
5. **Fuzzy Task:** many desired outcomes and multiple paths to achieve them (e.g., autonomous driving)

As complexity increases, the determinacy of a task decreases. Simple and problem tasks are suitable for full automation due to the fact that only one possible outcome exists. For simple tasks there is only one path to get to the desired outcome. For problem tasks, there is also only one desired outcome, but multiple possible solution paths exist that have to be computationally compared and evaluated. Due to the possibility of automation through exhaustive search, the authors claim there are no concerns about bias or inconsistent allocation. With decision tasks, the potential for full automation lowers. This represents a tipping point in the allocation logic. Multiple desired outcomes involve multiple paths leading to varied results that frequently entail conflicting interdependencies and trade-offs. Automation is still possible, but the intention of the goal has to be clearly stated, as achieving one outcome comes at the cost of another. However, determining which outcome should be prioritized involves subjective human reasoning that algorithms cannot inherently provide [29]. This relates directly to the concept of the intention layer by Xiang and Yu [29], where the human defines the goal and purpose of a task. Through a structural analysis, Vimalkumar et al. [26] identify that from the decision task level

onward, goals can compete. Xiang and Yu [29] add that resolving this competition requires human judgment at the intention layer, where the goal and the purpose of a task are defined. As will be discussed in Section 4.2.3, this remains a fundamentally human responsibility, as it is ultimately the human who decides what AI is used for and to what end.

The characteristics of judgment and fuzzy tasks further increase this complexity. Interdependencies between paths and probabilistic linkages between paths and outcomes make it harder to discern the actual path the system used to reach its solution. The models required to handle this complexity are often high-dimensional, which reduces the transparency of the decision-making process. This increasing complexity also correlates with the granularity of a task. Simple and problem tasks tend to be atomic in nature. They are well defined and self-contained, which allows for a direct allocation decision. As tasks move toward the fuzzy end of the spectrum, they become increasingly coarse-grained and encompass more sub-tasks with their own interdependencies. As established in Section 2.3, this means that before an allocation decision can be made, these complex tasks must first be decomposed. The more complex a task, the more decomposition is required, and the more the allocation problem shifts from a single decision to a series of decisions across multiple sub-tasks. The analysis of structural determinacy reveals a critical pattern. With increasing complexity, the limitations of a purely structural analysis become more apparent. However, the need to consider additional factors is not exclusive to complex tasks. Even a structurally simple task such as a basic calculation can become critical when it is embedded in a sensitive context, for example, when it determines the creditworthiness of a person. The structural logic of a task is therefore a necessary but never a sufficient basis for the allocation decision. It must always be complemented by an analysis of the contextual requirements (Section 4.3), the risk and accountability associated with potential failure (Section 4.4), and the dynamics of collaboration between human and AI agents (Section 4.5).

4.2.2 Cognitive Requirements

The previous section looked at the complexity of a task as a whole. A complementary approach is to break a task down into the mental steps it requires. This helps to see which parts of a task AI can take over and which parts still need a human. It also provides a way to decompose more complex tasks into smaller sub-tasks that can each be assessed individually. Stohr and O'Rourke [24] build on the foundation of “cognitive functions,” where the underlying idea is that “thinking can best be understood in terms of representational structures in the mind and computational procedures that operate on those structures.” Five cognitive functions were found to be relevant in their case study: decision-making, learning, perceiving, reasoning, and planning. A distinction is made between “core cognitive functions” (CCFs) and “symbiotic cognitive functions” (SCFs). CCFs are those functions where AI systems inherently outperform humans due to the inability of humans to perform those functions to the desired extent. Tasks that rely on CCFs offer high automation opportunities, with the possibility of substituting human

labor. In the context of predictive maintenance, functions like perceiving (gathering data via sensors) and reasoning (inferring causal conclusions from massive datasets) possess a high degree of mathematical logic and are considered CCFs due to machine superiority in those tasks. Conversely, symbiotic cognitive functions lack the deterministic logic required for full substitution. These functions involve ambiguity and subjective preferences, making the allocation problem more complex. For SCFs, the functions have to be further decomposed or, rather, shared between human and AI actors. In the context of PdM, the planning and learning functions are shared between human and AI agents, where AI provides insights (e.g., machine failure dates), while the human contributes experience-based knowledge to form a strategy. An important note is that the decision of whether a task is a CCF or an SCF is context-dependent and is influenced by additional aspects such as nuanced context understanding. For example, perceiving can mean either sensor-based state assessment or detection of deep emotional nuances, depending on the set goal and context of a task. This confirms that while the structural logic identifies what can be substituted, it cannot account for the nuanced contextual understanding or ethical stakes required for a task. Schroder et al. [22] apply a similar logic in the context of digital services, supporting the view that this granular decomposition is applicable beyond predictive maintenance.

4.2.3 Equivalence Layers and Agency Shifts

While the previous section identified which cognitive components of a task can be handled by AI, this does not yet clarify at which point in the process the responsibility shifts between humans and AI. Xiang and Yu [29] address this by decomposing a task into four sequential layers, each representing a stage where agency can shift between agents. Every task starts with an intention where the goal is defined. This is followed by a reasoning layer, where the involved data is represented and analyzed. The subsequent decision layer is where a decision is made based on the results of the reasoning. The final action layer is where decisions are executed. According to the authors, the intention layer remains a human stronghold, as defining goals relies on subjective values that lack a deterministic ground truth. At the reasoning and action layers, machines can take a leading role through data processing and automated execution. However, even when reasoning is fully automated, the process often returns to the human at the decision layer, as decisions in knowledge work require the consideration of contextual and moral factors that go beyond the data.

4.2.4 Data Landscape and Data-Readiness

The complexity of a task also determines what kind of data is required to complete it. A simple task operates on well-defined and structured inputs, where the required data is clear. As tasks become more complex, they increasingly depend on unstructured, fragmented, or even tacit knowledge that is difficult to capture in digital form. The first challenge is data availability. A task cannot be algorithmically formalized if the necessary information is insufficient or inaccessible. In many organizations, critical knowledge exists

only as tacit expertise in the minds of employees or in analog forms that have never been digitized [13]. Even when data has been digitized, it is often fragmented across departments or located outside the organization, creating high search costs [13]. The second challenge is data quality. According to Yun et al. [30], 80% of organizational data in knowledge work is unstructured or messy. Business environments are often characterized by undocumented knowledge, vague process documentation, data silos, and outdated product information [13]. For AI to produce reliable outputs, this data must be systematically organized and made accessible. Additionally, the temporal relevance of the required data plays a role. Whether a task depends on real-time or periodic data further dictates the complexity of the technical architecture needed.

The data landscape of a task is therefore not an independent factor but a direct consequence of its structural complexity. The more complex a task, the more its data requirements extend beyond structured and accessible sources into the territory of tacit and fragmented knowledge. This analysis of the structural logic of a task, from its composition and cognitive requirements to the data it depends on, provides a foundation for assessing technical feasibility. However, structural feasibility alone does not determine whether a task should be automated.

4.3 Interpretive and Contextual Nuance (Context)

The second decision factor for task allocation shifts the focus from structural feasibility to contextual understanding of the task environment. While the previous decision factor assessed whether task steps are logically formalizable, the context factor evaluates the degree to which a task requires an understanding of unspoken rules, social dynamics, and nuanced contextual understanding of a specific business or cultural environment. Contextual nuance refers to the subtle details within a specific situation that are often not explicitly stated but implicitly shape the quality of the outcome. These details are difficult to formalize because they are rarely captured in structured data. AI is limited to patterns within its training data, whereas knowledge work often involves variables that are unstructured, context-dependent, and difficult to anticipate.

4.3.1 The Importance of Nuanced Context Understanding

The previous section established that complex tasks increasingly depend on tacit knowledge. This section examines what happens when AI lacks access to this knowledge. The core problem is contextual blindness. Some human characteristics are lost when fully automating tasks. This decision factor, among other things, assesses whether this translation loss is bearable. Humans are generally considered better at tasks that require empathy, emotional intelligence, or moral judgment. When these inherent human characteristics are fundamental to the output value of a task, then full automation cannot be achieved yet. Those parts of the task that heavily rely on those human characteristics remain under human control.

A definitive example of the importance of this decision factor is found in Plant Jammer, a Danish digital service company investigated by Schroder et al. [22]. Plant Jammer offers a platform that creates personalized vegetarian recipes using a neural network trained on over 3 million recipes. The main goal is to replace meat with vegetarian ingredients only. From the perspective of the first decision factor (Section 4.2), this task is highly formalizable. The AI can calculate over 1.7 billion flavor combinations based on statistical frequency [22]. However, the key decision factor of Interpretive and Contextual Nuance shows that AI suffers from contextual blindness. In the case of Plant Jammer, the AI model identifies that two ingredients frequently appear together in the data, therefore concluding that they work well together. The problem is that the model lacks the sensory reality required to judge whether a combination is actually delicious or culinarily balanced. To overcome this blindness, Plant Jammer works together with food scientists, who developed a framework based on the five basic tastes (sweet, sour, salty, bitter, umami). In this configuration, the human becomes the contextual anchor. The AI suggests ingredients based on its data, but the human frames these suggestions by manually tagging ingredients and preparation methods. This ensures the output aligns with human sensory expectations. Schroder et al. [22] describe this as direct supervision. The human contributes tacit knowledge regarding aesthetics and flavor profiles that the AI, despite its 3 million data points, cannot replicate. So, the real challenge for this specific task lies in the nuance. How do you replace the mouthfeel or the umami of meat in a complex vegetarian dish? This requires intuition and creative problem-solving. If an organization ignores this nuanced complexity in a context where it is truly important for the value creation, it risks producing poor outcomes. The results are soulless, logically correct recipes that are culinarily worthless.

The study of Yun et al. [30] shows a similar pattern in a different domain. In their study of the system Yodeai, knowledge workers often felt a disconnect from AI-generated insights, perceiving them as too verbose or lacking personal connection. Generative AI excels at identifying popular topics and organizing hundreds of documents. However, humans are significantly more adept at understanding the emotional nuance of specific wording. A product manager reading a transcript perceives the frustration, hesitation, or excitement in a customer’s phrasing. AI, operating on statistical probability rather than lived experience, lacks this ability. Its summaries can feel shallow to the professional who relies on that emotional nuance to make evidence-based decisions.

Xiang and Yu [29] provide a more general explanation for why this problem arises. In their model, every task begins with an intention layer where the goal is defined. The machine then reasons and acts based on that goal. This creates a dependency. If the goal changes or was not clearly defined in the first place, everything the machine produces afterward is built on a wrong foundation. The authors illustrate this with the example of an intelligent healthcare team. A doctor defines the treatment goal for a patient, and the machine analyzes historical cases and literature to infer possible treatment plans. If the doctor adjusts the treatment goal based on a conversation with the patient, for example, shifting from aggressive treatment to palliative care, the machine’s previous reasoning

becomes irrelevant. It was optimizing for the wrong objective. In knowledge work, this kind of goal shift happens frequently because objectives are shaped by human context that evolves over time. This suggests that contextual nuance becomes a critical factor in the allocation decision whenever the problem definition is not stable or not explicitly contained within the data. Both examples illustrate this fundamental problem. A task can be structurally suited for automation and still produce inadequate outcomes if the context requires a form of understanding that cannot be derived from data alone.

4.3.2 The Relationship Dimension: Social Capital as Business Value

Beyond interpretation, Engel et al. [13] identify the Relationship Dimension as a core decision factor of allocation. This aspect is included in the interpretive and contextual nuance decision factor because navigating professional relationships relies heavily on tacit knowledge regarding social norms, emotional subtext, and unwritten communication rules. In the model proposed by Engel et al. [13], a high requirement for social presence directly correlates with lower amenability for cognitive automation. This factor considers that many professional tasks are not merely about information processing but about social capital and trust-building. In industries where the human touch is a cornerstone of business value, such as high-end consulting or medical praxis, full automation is impractical because removing the human element risks reducing a specialized service to a generic commodity. Whether this limitation remains valid or will be challenged by increasingly capable AI systems is discussed in Chapter 5. Both the contextual blindness discussed in the previous section and the relationship dimension point to the same conclusion. There are aspects of knowledge work that depend on inherently human qualities that cannot yet be captured in data or simulated by algorithms. However, context alone does not determine whether a task should remain under human control. The consequences of getting it wrong matter equally. The following section examines this dimension of risk and accountability.

4.4 Risk and Accountability

The previous sections assessed whether a task can be automated based on its structural logic and whether its context allows for meaningful automation. This section addresses a different question. If the automation produces a wrong output, what are the consequences? The level of risk associated with a task determines how much human control must be maintained, regardless of what is structurally or contextually possible. A task that is both simple in structure and free of contextual nuance may still require human oversight if the consequences of failure are severe.

4.4.1 Risk as a Decision Factor for Human Control

Bidar et al. [4] approach the allocation problem from a risk-informed perspective. Their research focuses on the public sector, where citizens are bound to government services and cannot simply choose an alternative provider. This makes the consequences of wrong

AI decisions particularly significant. The importance of this perspective is illustrated by incidents such as the Dutch “toeslagen-affaire,” where an over-reliance on automated decision-making led to serious harm for thousands of families [4]. The authors classify tasks along two dimensions: the degree of specialization and the level of risk. While the degree of specialization is covered by the first decision factor (Section 4.2), the level of risk serves as the basis for this section. Risk is defined as the potential harm of the AI application to stakeholders. Limited risk refers to tasks with a low probability of direct harm, where potential damages are recoverable. Significant risk refers to tasks with direct influence on stakeholders’ well-being or safety. From this classification, two task types are particularly relevant. High-reliability tasks require low levels of expertise but pose high risk to stakeholders. Wicked tasks require high levels of expertise and pose high risk. The central insight is that the criticality of a decision determines the amount of human control needed. Full automation is only justifiable if the potential risk of a wrong decision is low and leads to no significant impact. This also serves as an assurance of accountability. When a human remains in the loop, responsibility for the outcome can be clearly assigned.

4.4.2 The Complexity-Opacity-Fairness Chain

While Bidar et al. [4] assess risk from the perspective of stakeholder harm, Vimalkumar et al. [26] identify a mechanism through which risk emerges from within the technology itself. Their argument builds on the structural complexity discussed in Section 4.2. As tasks move from simple to fuzzy, the mathematical models required to solve them become increasingly high-dimensional. This leads to what is commonly referred to as the black-box character of AI, where the internal logic of the model becomes incomprehensible to humans. Vimalkumar et al. [26] use the theory of algorithmic opacity by Burrell [5], who identified three sources. The first is opacity from intentional corporate or state secrecy, where organizations keep their processes hidden to protect competitive advantages. The second is opacity requiring specialized technical literacy, where solutions are so sophisticated that understanding them requires a certain level of expertise. This form of opacity can be addressed through explanation. The third is opacity from high-dimensional optimization, where the sheer number of parameters in a model exceeds human capacity to comprehend. This is the form of opacity that poses the greatest challenge, because it cannot be resolved through explanation alone.

The consequence for the allocation decision is twofold. First, as complexity increases, so does procedural opacity, making it harder to verify whether the process that led to a decision was fair. Second, as tasks involve multiple competing outcomes, the risk of biased results increases. To assess algorithmic fairness, Vimalkumar et al. [26] draw on two fundamental legal doctrines: distributive fairness, which concerns the fair allocation of outcomes, and procedural fairness, which concerns the fairness of the process used to reach those outcomes. In high-stakes judgment or fuzzy tasks, the risk of hidden bias necessitates a shift toward augmentation, where a human acts as a moral anchor who takes responsibility for the outcome.

4.4.3 Accountability and the decision threshold

Both perspectives converge on the same conclusion. When the stakes are high, the allocation logic must ensure that a human retains control at the point where accountability resides. Building on the equivalence layers discussed in Section 4.2.3, this means that while AI can safely handle the reasoning layer for a high-risk task, the decision and action layers represent a threshold where legal and ethical accountability must remain with a human. Even if the AI’s reasoning is faster and more comprehensive, the final decision in a high-stakes context cannot be delegated to a system that cannot be held responsible for its outcomes.

The “toeslagen-affaire” introduced in Section 4.4.1 illustrates what happens when this threshold is not respected. Because this process was largely automated and lacked meaningful human oversight at the decision layer, two things were lost simultaneously. First, the ability to account for individual circumstances that the data does not capture. Second, a clearly accountable human who can be held responsible for the decision.

4.5 Multi-Agent Collaboration Dynamics

The previous three decision factors approach the allocation problem from the perspective of a single task, analyzing its structural logic, its contextual requirements, and the risk of potential failure. However, a central finding of this synthesis is that tasks in knowledge work are rarely executed by a single agent. Often, the allocation strategy is not a binary choice between full human or full AI execution, but rather a decomposition of the task into sub-tasks, each assigned to the agent best suited for it. Bidar et al. [4] illustrate this concretely in their four proposed case studies, where a single use case such as a government service process is broken down into multiple sub-tasks, executed by both human and machine actors together. A critical observation here is that this redistribution of sub-tasks does not only divide existing work; it reshapes roles entirely. Tasks previously owned end-to-end by a human professional may be restructured such that the human transitions from executor to supervisor, validator, or exception handler. While this decomposition resolves the question of individual task-level feasibility, it simultaneously introduces new challenges. Once multiple agents (human and machine) work together to produce a shared output, the interaction between those agents becomes a decision factor for the allocation decision in its own right. Xiang and Yu [29] explicitly identify this as one of the core challenges of human-machine hybrid intelligence. Multi-agent collaboration faces numerous potential frictions, including trust, privacy, and coordination, that do not exist when a single agent executes a task in isolation. This fourth decision factor therefore evaluates the collaboration overhead introduced by a given decomposition and asks whether the resulting multi-agent team is worth the resulting overhead.

4.5.1 Trust Calibration between Agents

Xiang and Yu [29] identify trust as one of the core issues in multi-agent cooperation, and Dubois and Le Ny [11] model it as a dynamic stochastic variable that rises after successful AI interactions and falls after failures. In their framework, trust describes the degree to which a human operator accepts and acts upon the outputs of an AI agent as a valid basis for their own actions [11]. Trust is the critical human variable that can override any structural or risk-based allocation decision. Even if a sub-task is assigned to the AI agent on the basis of the first three decision factors, the human agent's trust state at any given moment may cause them to override, second-guess, or ignore the AI's output, effectively nullifying the allocation decision. A decomposition that looks appropriate on paper may therefore break down in practice if the trust relationship between agents is not actively maintained. Two failure modes emerge from this dynamic. The first is over-trust, where the human places too much confidence in the AI system and stops thinking critically about its outputs, even in those parts of the task that were kept under human control. Yun et al. [30] observed this in their study of knowledge workers who, over time, stopped independently evaluating AI-generated summaries, even in cases where their own judgment was supposed to be the final check. This is not a problem with automation itself but with automation creeping into areas where human oversight was still required. Xiang and Yu [29] illustrate the most severe consequences of this failure mode with the 2019 crash of the Ethiopian Airlines Boeing 737 MAX 8, where pilots relied on the automated system and failed to take manual control even as the machine's judgment was demonstrably flawed. The inverse failure mode is under-trust, where the human underestimates the AI system's actual capabilities and, as a result, avoids delegating tasks to it [11]. Vimalkumar et al. [26] observe this in the context of domain experts, who tend to remain skeptical of AI outputs in high-complexity tasks because their own expertise makes them aware of how uncertain the problem really is. From an allocation perspective, under-trust has a practical consequence, where the human ends up doing tasks that the AI could have handled reliably. This increases the human's workload and brings back the kind of errors that come with repetitive manual processing. A third and longer-term risk is deskilling. When the AI takes over routine sub-tasks, the human stops practicing those tasks over time. As a result, the human gradually loses the ability to judge whether the AI's outputs are actually correct. If the AI then fails or encounters an unfamiliar situation, the human may simply no longer have the skills to take over [30]. Beyond individual task performance, deskilling also raises a broader concern about organizational resilience. In critical sectors such as healthcare or infrastructure management, over-reliance on AI systems creates a structural vulnerability. If those systems become unavailable, for example, due to technical failures or power outages, the organization may lack the human competence required to maintain operations manually. This suggests that deskilling is not only a quality concern but also a risk that needs to be considered in the allocation decision. Engel et al. [13] implicitly address this through their emphasis on iterative development and continuous human involvement, ensuring that domain expertise is not fully outsourced to the AI system.

4.5.2 Coordination and Communication Between Agents

Beyond trust, Xiang and Yu [29] identify coordination as a fundamental challenge in multi-agent systems. Each agent, whether human or machine, operates with its own goals, constraints, and decision logic. When these are not aligned, conflicts arise. In knowledge work, this can lead to unclear responsibilities, where no agent knows who owns a given sub-task. As a result, agents may duplicate work without realizing it, or a sub-task may be neglected entirely because each agent assumes the other has handled it. Zhang et al. [31] address this through their role-based allocation model, which evaluates agents not only by what they can do independently but also by how well they perform when working together with another agent. Their key insight is that some agent combinations produce better results than either agent could achieve alone, while others create so much friction that overall performance actually decreases. This means that when decomposing a task, it is not enough to ask what each agent can do. It is equally important to consider how cleanly the output of one agent can be handed over to the next. If the handover requires additional interpretation, background knowledge, or back-and-forth between agents, the coordination effort may outweigh the benefits of the decomposition in the first place.

4.5.3 Privacy and Data Integrity at Agent Interfaces

A further friction in multi-agent collaboration concerns privacy and data integrity, which Xiang and Yu [29] identify as a key challenge in multi-agent systems. When multiple agents work together, information has to flow between them. This creates points of exchange that are vulnerable to unauthorized access or data misuse. In knowledge work this is not only a technical problem but also an organizational and legal one. Sub-tasks assigned to AI agents often require access to sensitive client data, internal business information, or personal records. Engel et al. [13] identify data silos and inaccessibility as a direct obstacle to cognitive automation, because the data needed to execute a sub-task reliably is often fragmented across departments, poorly documented, or restricted. The allocation decision must therefore also consider whether the necessary data can actually be made available to the AI agent in a way that is both technically functional and legally compliant.

4.5.4 Collaboration Overhead as an Allocation Constraint

Taken together, the three frictions (trust miscalibration, coordination costs, and data interface risks) produce a collaboration overhead that must be weighed against the efficiency gains expected from decomposing a task. A decomposition that seems justified based on the first three decision factors may still be a poor choice if the resulting collaboration between agents generates more problems than it solves. This is the core contribution of the fourth decision factor. It does not ask what individual agents are capable of, but whether they can work together effectively enough to make the decomposition worthwhile. As Xiang and Yu [29] conclude, addressing these challenges

requires consideration of factors from technology, law, ethics, and society, going well beyond the structural analysis of any individual task. This concludes the identification and analysis of the four key decision factors. The following section builds on these findings and translates them into practical design guidelines.

4.6 Applying Principles to Work (Re)Design

The previous sections identified four key decision factors that collectively inform the allocation decision: the structural logic of a task, interpretive and contextual nuance, risk and accountability, and multi-agent collaboration dynamics. This section translates these analytical findings into actionable design guidelines that address the second research question (RQ2): how the identified principles can be applied to help (re)design work processes for responsible and reliable AI integration.

4.6.1 Defining Process Intention

Before technical design decisions on the process can be made, the intention behind the process must be clearly defined. This prerequisite is necessary because it determines how the four decision factors are interpreted and design trade-offs are made. As proposed by Xiang and Yu [29], every task begins with an intention layer, where the goal is defined. This idea can serve as a primary driver for the design of the entire system. The reason is that the same set of decision factors can lead to fundamentally different design decisions depending on the process intention. Two processes that are similar in the four identified key decision factors should be designed differently based on the intention. If the intention of the first process is scaling, as in the case of Plant Jammer [22], the design will prioritize throughput and consistency. The AI is given computationally intensive sub-tasks (calculating flavor combinations at scale), while a small team of human experts provides contextual and tacit knowledge to achieve high-quality outputs. The design accepts a degree of contextual loss in exchange for the ability to serve a large user base. The contextual loss is actively addressed through the small expert team of humans. If the intention of the second process is risk minimization, as in the public sector use cases described by Bidar et al. [4], the same structural complexity leads to a fundamentally different design. Here, throughput is secondary to safety. The design must ensure that every critical decision passes through a human checkpoint, even if this slows down the process significantly.

Therefore, the first step in any process (re)design is to articulate clearly what the process is meant to achieve and for whom. Only when this intention is defined can the decision factors be meaningfully applied.

4.6.2 Degree of Automation

This section addresses the core allocation decision: given the four decision factors analyzed, what degree of automation is appropriate for the given task? After clearly defining the

intention of a process, the degree of automation can be decided more systematically. This simultaneously determines the degree of human control. The four decision factors collectively inform this decision, but they do so in different directions. Structural logic may suggest that a task is fully automatable, while contextual nuance or risk considerations may require that a human remain involved. The equivalence layers from Xiang and Yu [29] provide a structural framework for placing intervention points within any given mode. As established in Section 4.2.3, the intention layer remains a human responsibility across all modes. The reasoning and action layers offer the highest automation potential. The decision layer represents the critical threshold where, depending on the risk profile, human control must be preserved. This framework allows one to define how much autonomy the AI receives and also at which cognitive layer.

Full automation is possible if it is a simple or problem task where contextual nuances are not required, the potential risks are low, and the overhead of agent collaboration is low as well. Tasks that are repetitive, data-intensive, and well-defined fall into this category. Schroder et al. [22] argue that such tasks should be primarily handled by AI to ensure consistency and accuracy. In terms of control architecture, the AI operates autonomously across all four equivalence layers. The human defines the intention at design time but does not intervene during execution. Human involvement is limited to periodic review and the option for stakeholders to escalate to a human if they believe the AI's output is incorrect [4]. Bidar et al. [4] classify such kinds of tasks as “mundane,” which require a low degree of specialization and pose little to no risk. Humans are freed from the simple, repetitive parts of a task.

AI-led automation (Human-on-the-loop) is suggested if the task is structurally automatable, but the risk profile or contextual requirements demand that a human remain available to intervene. This mode is appropriate for tasks at the decision level of the hierarchy proposed by Vimalkumar et al. [26], where multiple outcomes exist and trade-offs must be managed, but where the risk remains moderate. It also applies to the high-reliability tasks described by Bidar et al. [4]: structurally simple but with significant potential harm to stakeholders.

In this mode, the AI takes a leading role at the reasoning and action layers. It processes data, generates recommendations, and can execute routine decisions autonomously. However, the human monitors the process at the decision layer and intervenes when anomalies or escalations occur. The critical design feature is the escalation mechanism: the system must be designed to detect conditions under which human intervention is needed and to route those cases reliably. In the high-reliability task design proposed by Bidar et al. [4], the repetitive workload was similarly delegated to AI, but a separate manual process was preserved in parallel, and humans maintained an active supervision role. The key insight is that in this mode, the human's value lies not in processing every case but in catching the cases that the AI gets wrong.

Human-led operation with AI assistance becomes necessary when high potential risks or high contextual nuance is involved. This applies to judgment and fuzzy tasks [26], where the structural logic of a task often demands advanced AI models. The high-risk and/or

highly nuanced context requires human oversight due to the opacity of those algorithms and the context itself. In this mode, the human retains agency at both the intention and decision layers. The AI contributes at the reasoning layer by processing large volumes of data, identifying patterns, and generating proposals that reduce the human's cognitive load. However, the final decision and the subsequent action remain under human control.

In the expert task design proposed by Bidar et al. [4], this was implemented through a combination of a random forest model (for interpretable reasoning) and a deep neural network (for performance), paired with a GUI that presented the AI's results in a way that supported the expert's own decision-making. The human experts were also required to collect more nuanced data from customer interactions, establishing a feedback loop that continuously improved the AI's reasoning layer. The wicked tasks described by Bidar et al. [4] represent the most demanding form of this mode. Here, the core process remains entirely human-led, enriched with a data-driven component that supports but never overrides human judgment. The AI's autonomy is strictly limited, extensive ethical oversight is integrated, and in some cases, an entirely new human role is created whose primary function is to collaborate with the AI: classifying its outputs and incorporating them into their own decision-making. The design effort required for this mode is significantly higher than for the other two, reflecting the severity of the consequences if the system fails.

An important caveat is that the selection of the automation mode is not necessarily a one-time decision. As Dubois and Le Ny [11] demonstrate, the optimal allocation between human and AI can shift dynamically based on the human operator's current trust and cognitive workload. A system designed for AI-led augmentation may need to temporarily shift toward human-led operation when trust drops below a critical threshold or when the human's workload allows for deeper engagement. This dynamic dimension of trust is addressed in Section 4.6.4.

4.6.3 Explainability and Transparency

Once the automation mode is selected, the design must address how transparent the AI's reasoning needs to be. Section 4.4.2 established that as task complexity increases, the models become more opaque, which in turn threatens fairness. However, not all tasks need full transparency. The level of explainability required depends on the four identified decision factors. For tasks with low complexity and low risk, where the automation mode is full automation, performance can take precedence over explainability. If the task requires a low level of human expertise, the outputs are easier to verify. In the high-reliability tasks described by Bidar et al. [4], a self-service approach was used where the model's performance was prioritized over its interpretability because the context required fast processing and the outputs could be verified through their results rather than their reasoning.

For high-complexity, high-risk tasks, where humans have a decision-making or validation role, explainability becomes a prerequisite. If the human cannot understand the basis of

the AI recommendation, the human-in-the-loop architecture becomes more of a formality than a genuine safeguard. To address this, Bidar et al. [4], in the expert task use case (high complexity, high risk), paired a random forest with a deep neural network. The idea behind this is that the random forest provides interpretable decision factors, while the neural network contributes performance. This shows that at the algorithmic level, the choice of the machine learning model directly influences the inherent interpretability. Interpretable models like random forests or logistic regression offer transparency by design, whereas deep learning models require post-hoc explanation techniques to approximate interpretability [4]. At the interface level, graphical user interfaces (GUIs) that present AI outputs in a way that supports human reasoning are essential for any mode that involves human oversight [4]. The GUI is not merely a presentation layer but a design tool that shapes how the human perceives and evaluates the AI contribution. Without it, even an interpretable model output may remain inaccessible to a non-technical expert. Vimalkumar et al. [26] propose specific affordances that address different aspects of transparency. These include making the process visible to the user (transparency), translating complex outputs into compact visualizations (visualization), allowing users to see alternative outcomes and compare them (comparability), and giving affected stakeholders a mechanism to challenge or question automated decisions (voice).

As established in Section 4.4.2, the increasing complexity of a model directly threatens both distributive and procedural fairness. When the internal logic of a model is incomprehensible, it becomes impossible to verify whether the process that led to a decision was fair and difficult to detect whether outcomes are systematically skewed. This means that beyond explainability, design choices have to include active monitoring of outcomes for bias. This includes regular audits of the AI outputs, systematic comparison of outcomes across different demographic or stakeholder groups, and documented review processes that create an accountability trail. Bidar et al. [4] reinforce this by arguing for extensive ethical oversight in wicked task scenarios, including alignment with regulatory frameworks such as the European AI Act. When the intention is efficiency or scaling and the risk is low, performance takes precedence. When the intention is compliance, stakeholder protection, or any context where the consequences of failure are significant, explainability and fairness take precedence, even at the cost of throughput. This is not a technical limitation but a design choice that reflects the values and priorities embedded in the process intention.

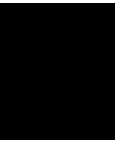
4.6.4 Managing Trust and Redesigning Roles

The fourth decision factor (Section 4.5.1) shows the role that trust between human and AI agents plays. Design choices need to address trust issues like over-reliance and under-reliance, but also other problems like deskilling. Strategies to prevent over-reliance include cognitive forcing functions that force the user to actively engage with AI recommendations before accepting them [6]. This mechanism aims to keep the human cognitively engaged. However, Buginca et al. [6] note a trade-off involving subjective experience, where users reported a less favorable user experience.

One option to prevent under-reliance is explainability. When the user sees the reasoning behind the AI decision, they can compare it with their own judgment and better comprehend the outputs. Dubois and Le Ny [11] model this explicitly through trust-seeking behavior, in which the AI performs well on observable tasks, gradually building the human's confidence in its abilities.

To directly address the problem of deskilling, mechanisms to maintain human competence are required. Crowston and Bolici [10] propose a model showing that whether an AI system leads to deskilling or upskilling depends on the opportunities for users to edit and evaluate AI recommendations. Design mechanisms include periodic manual execution of automated tasks and rotation policies to ensure human capabilities are maintained [10]. Engel et al. [13] additionally support this by iterative development and continuous human involvement.

Beyond maintaining human competencies, the introduction of AI into work processes fundamentally reshapes human roles. Bidar et al. [4] show that the role of the human agents in the process redesign directly depends on the task type. In mundane and high-reliability tasks, the human role shifts to supervisor and validator, monitoring automated inputs and intervening when escalations occur. In highly complex, low-risk tasks, the human, in addition to their supervision role, collects more nuanced data to improve the model. In highly complex, high-risk tasks, an entirely new human role may be required, whose function is to collaborate with the AI and incorporate them into their own decision-making.



Discussion

The analysis presented in this thesis identifies four key decision factors for task allocation and derives a set of design guidelines from them. Before concluding, it is worth reflecting critically on what these findings actually achieve and where they fall short.

A first observation concerns the empirical grounding of some of the central claims in the reviewed literature. Several arguments rest on assumptions that are difficult to falsify. The claim that humans are better at empathy, emotional judgment, and contextual reasoning is treated as given across much of the literature. However, this is a normative assumption rather than an empirically proven fact. Research in behavioral economics and cognitive psychology has shown that human judgment is inconsistent and subject to well-documented biases. A systematic review of decision fatigue in healthcare found that 45% of studies measuring the effect reported significant changes in medical decision-making over the course of a shift, with physicians becoming more likely to choose cognitively easier options as workload accumulated. Doctors and nurses often make different medical choices at the end of a long shift because their brains are tired [19]. If the argument for keeping humans in the loop rests on the superiority of human judgment, that argument requires stronger empirical support than the reviewed literature provides.

The main analysis established that AI currently struggles with tasks that require nuanced contextual understanding. This applies especially to tasks that require sensory grounding, emotional understanding, or the ability to respond to shifting human goals. The decision factors treat the ability to handle shifting goals and the role of human relational authenticity as core task values. The core argument that AI cannot read and integrate the relevant information that a human would naturally draw on is probably only a temporary constraint, as AI solutions evolve. Liu et al. [18] addressed a concrete form of contextual blindness, where the performance of large language models decreases significantly when relevant information is located in the middle of a long input context. Methods like retrieval augmented generation (RAG) address this problem and serve as a workaround [15]. Context window limitations are real, but they are addressed as

engineering problems rather than accepted as permanent. Treating them as a major argument against AI would be weak because, as technology evolves, this problem becomes smaller and smaller.

The stronger version of the argument revolves around whether relevant tacit knowledge can be transferred to the AI system at all. Autor [2] applied this directly to automation, arguing that tasks requiring adaptability, contextual judgment, and common sense remain resistant to machine substitution precisely because the underlying knowledge cannot be codified. The Plant Jammer case illustrates this well. The human food scientists were not retained because the AI lacked computing power. They were retained because the knowledge required to judge whether a recipe was actually delicious had no structured form that could be transferred to the system. In some cases, this limitation can be overcome through richer training data, user feedback loops, or structured domain knowledge. In those cases, the limitation is temporary. In other cases, it cannot. Some knowledge is too closely tied to lived sensory experience, or too deeply embedded in human relationships, to be captured in a form a model can use. In those cases, the argument holds. The framework would benefit from making this distinction explicit. Contextual nuance is not one thing. It is a weak argument against automation where knowledge transfer is feasible and a strong one where it is not.

Even where the tacit knowledge argument holds today, the question remains whether it will hold tomorrow. The main analysis argued that AI companions do not disprove the relationship dimension because people who use them are settling for a substitute rather than a genuine alternative. This avoids a harder question. What happens when the substitute becomes good enough that users can no longer tell the difference? The rapid growth of AI companionship platforms, driven in part by rising loneliness, suggests that for many users this threshold has already been crossed in practice. If AI-mediated connection produces the same effects as human connection, the argument that human relationships are categorically superior becomes difficult to apply in real decisions, even if it remains true in theory. The framework does not currently provide guidance for when a capability threshold has been crossed.

The risk and accountability decision factor provides a relatively stable basis for allocation decisions. When an automated system fails in a high-stakes context, someone must be responsible for that failure. AI systems cannot carry legal or moral accountability. A human must remain in the loop to fulfill that role. The main analysis showed what happens when that role is removed. The “toeslagen-affaire” illustrates how removing human oversight does not eliminate risk. It eliminates the ability to catch and correct failures before they become irreversible. However, this argument has a time condition attached to it. It rests on the assumption that humans are more reliable than the AI systems they are supervising. That assumption is already being challenged. AI diagnostic tools in radiology and pathology have reached or exceeded specialist-level accuracy in controlled settings [25]. If a system becomes consistently more reliable than the human overseeing it, the justification for keeping that human in the loop becomes harder to defend on performance grounds. The accountability argument would still hold in a legal

and institutional sense. But its practical force weakens as AI reliability increases. The framework should acknowledge this openly rather than treating human oversight as an unconditional requirement.

The risk decision factor contains a tension that the reviewed literature does not fully address. Requiring human oversight assumes that a human is actually available as a meaningful alternative. This assumption breaks down in contexts of resource scarcity. Consider a healthcare system operating beyond capacity. The realistic alternative to an automated diagnosis is not a human doctor reviewing the output. It is a three-month waiting period. In that situation, withholding automation in the name of human oversight may itself cause harm. A patient whose condition goes undetected because the system insisted on human verification is not better protected. They are worse off. This tension becomes sharper when access is considered. If a capable AI diagnostic system can reach patients who currently have no realistic access to specialist care, the argument for slowing deployment to preserve human oversight requires a direct answer to the question of what those patients are supposed to do in the meantime. The framework does not currently provide that answer. A more complete version would include criteria for when the costs of requiring human oversight outweigh its benefits, particularly in contexts where the human alternative is not genuinely available.

Agentic AI systems do not simply execute instructions. They decompose goals into sub-tasks, make decisions along the way, and take actions with real-world consequences across sequences of steps. This changes the nature of human oversight. It is no longer sufficient to have a human approve the final output. The decisions that matter may have been made earlier in the process, in ways that are difficult to trace or reverse [8]. The four decision factors were developed against a background assumption of relatively contained AI behavior. As systems become more autonomous, the framework would need to be extended rather than applied as is.

Beyond the technical question, highly capable AI systems require access to more data. In healthcare alone, maximizing diagnostic accuracy would require access to complete medical histories, real-time physiological monitoring, and behavioral data. The potential benefits are real. So are the risks. Data collected for one purpose can be repurposed against the interests of the people it was collected from. Existing algorithmic scoring systems already demonstrate this. These are not questions this thesis can resolve. They are political and societal questions as much as technical ones. What frameworks like this one can contribute is a clearer articulation of what is at stake in those decisions. That is a modest contribution. But given the pace of development, it is a necessary one.

Furthermore, the methodology of this thesis has its own constraints that should be acknowledged. The literature review is based on a corpus of nine core papers. While the selection followed a structured process, it is difficult to derive generalizable principles from such a small sample. The design guidelines derived for research question RQ2 are primarily based on the same corpus used for research question RQ1, supplemented by individual sources. A dedicated search for design-oriented literature could yield additional recommendations. The corpus includes publications from sources of varying academic

standing. A more exhaustive review with a larger corpus and a formal quality assessment scheme would strengthen the robustness of the findings.

Another point is that three of the nine core papers were identified through semantic discovery using the AI-supported platform SciSpace rather than through systematic database searches. While this approach helped broaden the scope of the review, the search logic of the tool is not fully transparent, which limits the reproducibility of this part of the search process. These methodological limitations do not invalidate the findings but define the boundaries within which the proposed framework should be interpreted.

Conclusion

This thesis set out to answer two research questions. The first asked which factors determine how tasks should be allocated between humans and AI in knowledge work. The second asked how those factors can be applied to (re)design work processes in practice. On the first question, four key decision factors were identified: structural logic, interpretive and contextual nuance, risk and accountability, and multi-agent collaboration dynamics. These do not operate independently. Effective allocation requires all four to be considered together. On the second question, five design guidelines were derived: defining process intention, selecting the degree of automation, calibrating explainability to risk, managing trust between agents, and redesigning human roles. Taken together, the framework offers a structured basis for making task allocation decisions that are grounded in the specific properties of a task rather than in general assumptions about what AI can or cannot do. The framework does not eliminate the difficulty of these decisions. It makes the relevant trade-offs visible and forces them to be addressed explicitly. However, as discussed in Chapter 5, it was developed for bounded AI deployments and does not yet account for agentic systems that define their own task boundaries during execution.

Future research should address the following directions. First, the design concerns identified in the thesis (explainability, fairness, trust, and accountability) represent additional requirements when human and AI actors are combined in a shared process. Current process modeling languages like BPMN and ArchiMate were not designed to express these concerns. Future work should investigate how these languages can be extended to capture allocation-relevant properties, such as the required degree of human oversight at a given decision point, the explainability requirements for an AI-supported task, or the trust state between collaborating agents. Before such extensions can be developed, the set of relevant design concerns must be clearly defined. This thesis provides a starting point by deriving several of those concerns from existing literature.

Second, the tension between accountability requirements and resource scarcity is already recognized in fields such as medical ethics and global health. When the realistic alternative

to an automated decision is not a human review but a three-month waiting period; withholding automation may itself cause harm. Future research on task allocation between human and AI agents should engage with this existing debate and develop criteria for when imperfect automation is preferable to no service at all. Third, the rapid development of agentic AI systems is already challenging the bounded assumptions of this framework. Emerging research on agentic AI should be evaluated for its implications for task allocation logic, particularly where the unit of analysis shifts from a predefined task to a goal and where human oversight must be designed into a process whose full shape is not known at the outset.

APPENDIX **A**

Synthesis Matrix

Table A.1: Synthesis matrix of core papers mapped against six analysis categories.

Author & Year	Task & Context	Task Factors	Human Factors	AI/System Factors	Allocation Logic	Design Principles
Yun et al. [30]	Knowledge work; processing, analyzing, and synthesizing information for decision-making	Domain knowledge reliance; unstructured data; iterative and collaborative processes	Deep contextual understanding; human verification; over-reliance risk; user isolation	Output quality limitations; data freshness challenges; contextual blind spots; bias and hallucinations	Complementary logic; AI provides initial support; humans retain validation and complex decisions	Adaptability and flexible workflows; accountability and transparency of AI outputs; interoperability and secure data integration
Xiang and Yu [29]	Task decomposed into identification, reasoning, decision, and action; hybrid augmented intelligence	Priority of tasks; deep contextual information needs; subjective perception and judgment; complex emotional situations	Problem definition and goal setting; subjective information processing; decision-making responsibility; supervision	Data processing and analysis; repetitive tasks; computational capabilities; prediction and recommendation	Humans for problem definition and goal setting; machines for data processing and analysis	Cognitive equivalence layers; transparent communication; encryption and data anonymization; task scheduling and resource allocation
Engel et al. [13]	Use cases as unit of analysis; cognitive automation; assessing CA appropriateness	Cognition requirements; complexity and volatility; ambiguity and novelty; interdependency; expert knowledge needs	Relationship requirements; transparency requirements	Data requirements; data silos; probabilistic outcomes; experimental character; black box characteristics	High amenability to CA indicates AI suitability	Use case deconstruction; stakeholder alignment; pre-investment and preparatory work; iterative development

Author & Year	Task & Context	Task Factors	Human Factors	AI/System Factors	Allocation Logic	Design Principles
Dubois and Le Ny [11]	Binary decision problem; trust and workload as dynamic variables	Binary nature; task frequency; objective difficulty	Mental workload as limit; trust as dynamic variable (rises with success, falls with failure)	Automation reliability; observability; transparency	Cognitive state variables; allocation based on current trust level	Trust seeking; trust repair
Zhang et al. [31]	Task allocation in human-machine collaboration ; multi-agent environment	Task decomposeability; resource requirements; task dynamics	independent capability profile; collaborative capability (emergent in HM-agent pairs)	Machine agent capability; capability evolution during collaboration	Role-based (RBHMC); short-board balance; capability heterogeneity	Synergy assessment; dynamic reconfiguration
Stohr and O'Rourke [24]	Cognitive functions lens; predictive maintenance; single cognitive function as unit	Core cognitive functions (CCF); symbiotic cognitive functions (SCF); data volume and velocity	Judgment; contextual reasoning; social intelligence	Type of cognitive function	CCFs assigned to AI; SCFs shared between human and AI	Allocation logic based on cognitive function type
Schroder et al. [22]	Digital services; task as part goal in larger process; Mintzberg coordination mechanisms	Service oriented; creativity vs. rationality	Standardization of norms	AI not able to work fully autonomously yet; augmentation needed initially	Control-cost-differentiation framework; service improvement, development, personalization	Cost efficiency, differentiation, and control allocation

Author & Year	Task & Context	Task Factors	Human Factors	AI/System Factors	Allocation Logic	Design Principles
Bidar et al. [4]	Decision-making tasks impacting stakeholders; risk-centered approach	Stakeholder risk (consequences of failure); task complexity and specialization level	Accountability and legal/moral responsibility; ethical imperatives	Safety-critical performance; reliability in high-stakes environments	Four-quadrant matrix: mundane, high-reliability, expert, wicked; driven by failure potential	Risk-based filtering; human-led for high-risk tasks; efficiency secondary to accountability
Vimalkumar et al. [26]	General knowledge work and information processing; task complexity and TTF	Intrinsic task nature (complicated vs. complex); task opacity; information asymmetry	Self-efficacy; perceived cognitive load; psychological trust based on perceived complexity	Ability to reduce information asymmetry through transparent outputs	Managing cognitive load; trust calibration;	Prioritize transparency for trust; reduce cognitive strain through intuitive interfaces

Overview of Generative AI Tools Used

For the purpose of improving readability and checking for spelling and grammar mistakes as well as for translating, the following AI tools have been used:

- Anthropic Claude Opus 4.7 and 4.6 for checking grammar and spelling, rewording phrases, suggesting synonyms, translating from German to English, and creating LaTeX table structures.
- Google Gemini 3 Pro and Gemini 3 Flash for checking grammar and spelling, rewording phrases, suggesting synonyms, translating from German to English, and creating LaTeX table structures.
- SciSpace for semantic search, topic discovery, and structuring and summarizing sources during the literature search phase.
- LanguageTool ([language-tool.org](https://www.language-tool.org)) for spell and grammar checking.

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